

Total No. of Questions : 8]

SEAT No. :

P2363

[Total No. of Pages : 2

[5870]-1531

T.E. (Computer Engg.)

DESIGN AND ANALYSIS OF ALGORITHM

(2015 Pattern) (Semester - II) (310250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q. 2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Make suitable assumptions wherever necessary.

Q1) a) Explain Time and Space complexity of an algorithm with example. **[5]**

b) What is the application of proof techniques? Explain any one technique with example. **[7]**

c) Solve following multiplication using Divide and Conquer strategy.
 $12345678 * 21394276$. **[8]**

OR

Q2) a) Discuss the advantages & disadvantages of recursion. **[6]**

b) Prove by induction that $S_n = 1 + 2 + 3 + 4 + \dots + n = n*(n+1)/2$ **[6]**

c) For given two sequences, Find LCS,
String S = "ABACCD", String T = "ACDF" **[8]**

Q3) a) What is Greedy algorithmic strategy? Explain Greedy approach for finding shortest path in weighted graph. **[8]**

b) Explain deterministic and non-deterministic algorithm with example. **[8]**

OR

P.T.O.

Q4) a) Explain asymptotic notations Big-O, Theta, Omega with example of each. **[9]**

b) Whether Travelling Salesman Problem is NP Hard problem or not? Justify. **[7]**

Q5) a) Explain amortized analysis of splay tree operations. **[9]**

b) Explain Randomized algorithm for Quick Sort. **[8]**

OR

Q6) a) Explain amortized analysis for Binary and Binomial Heap. **[8]**

b) Explain approximate algorithm for Vertex Cover. **[9]**

Q7) a) Explain Rabin- Karp string matching algorithm with example. **[10]**

b) Explain Distributed Breadth first search algorithm **[7]**

OR

Q8) a) Explain multithreaded Merge sort algorithm. **[9]**

b) Explain Naïve string matching algorithm with its best, average & worst case complexity. **[8]**

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